MILITARY SPECIFICATION SHEET

CIRCUIT BREAKERS, AIR, ELECTRIC, INSULATED HOUSING (SHIPBOARD USE) TYPE AQB - A50 AND NQB - A50

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation: MIL-C-17361.

AMSC N/A FSC 5925

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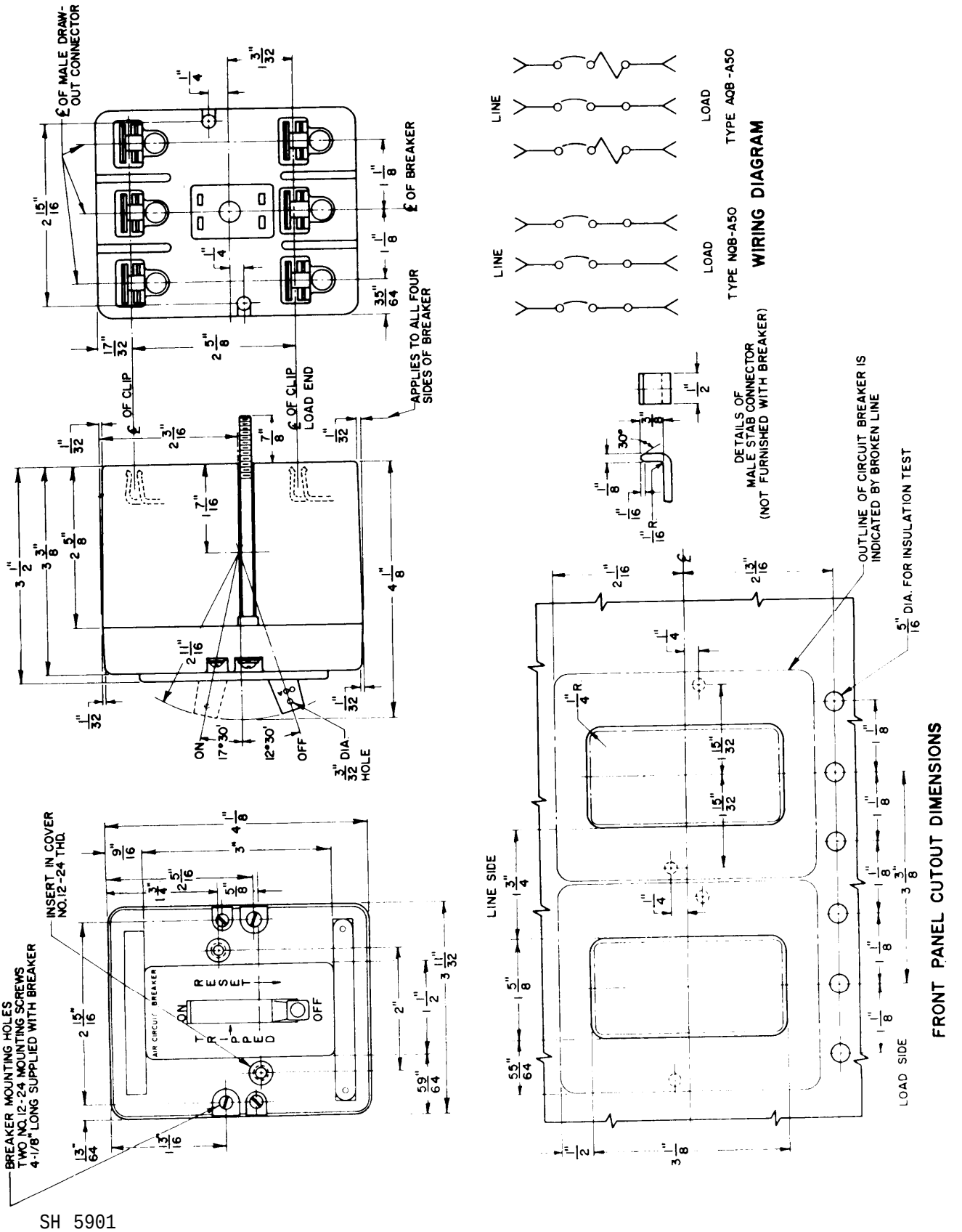
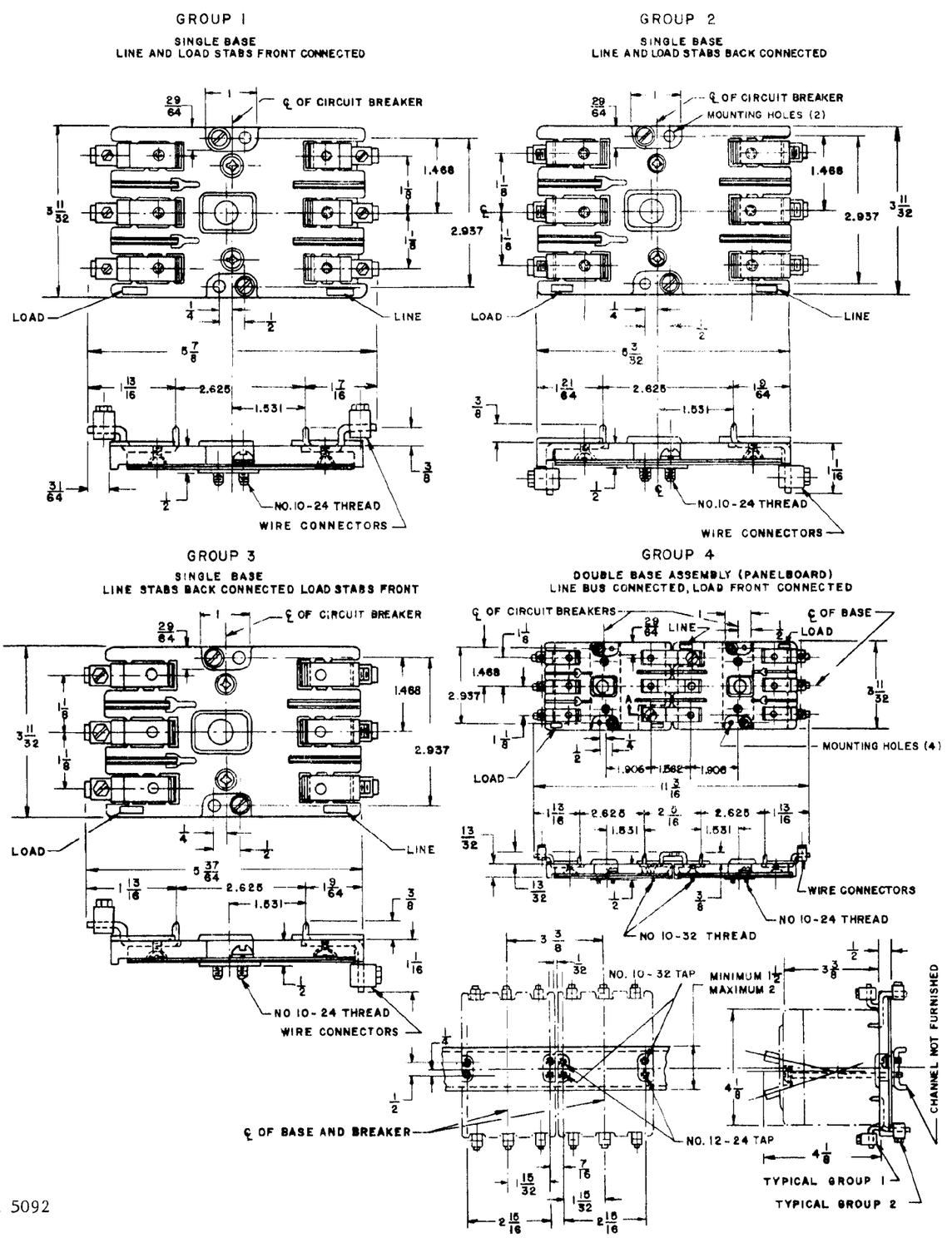


FIGURE 1. Circuit breaker type AQB-A50 and NQB-50 outline and mounting dimensions.

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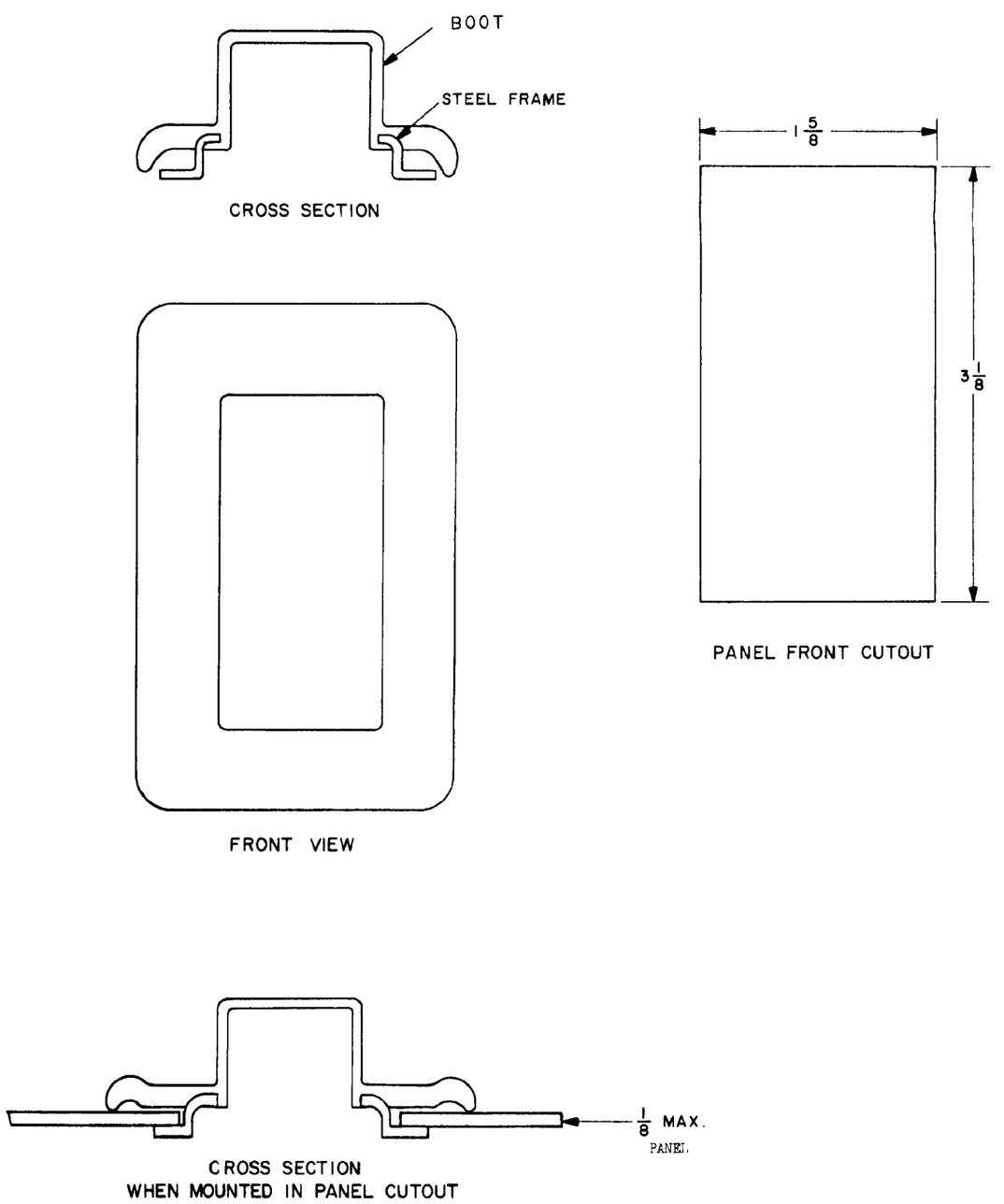
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FIGURE 2. Mounting bases for circuit breaker type AQB-A50 and NQB-A50 outline and mounting dimensions.

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FIGURE 3. Circuit breaker handle boot for AQB-A50.

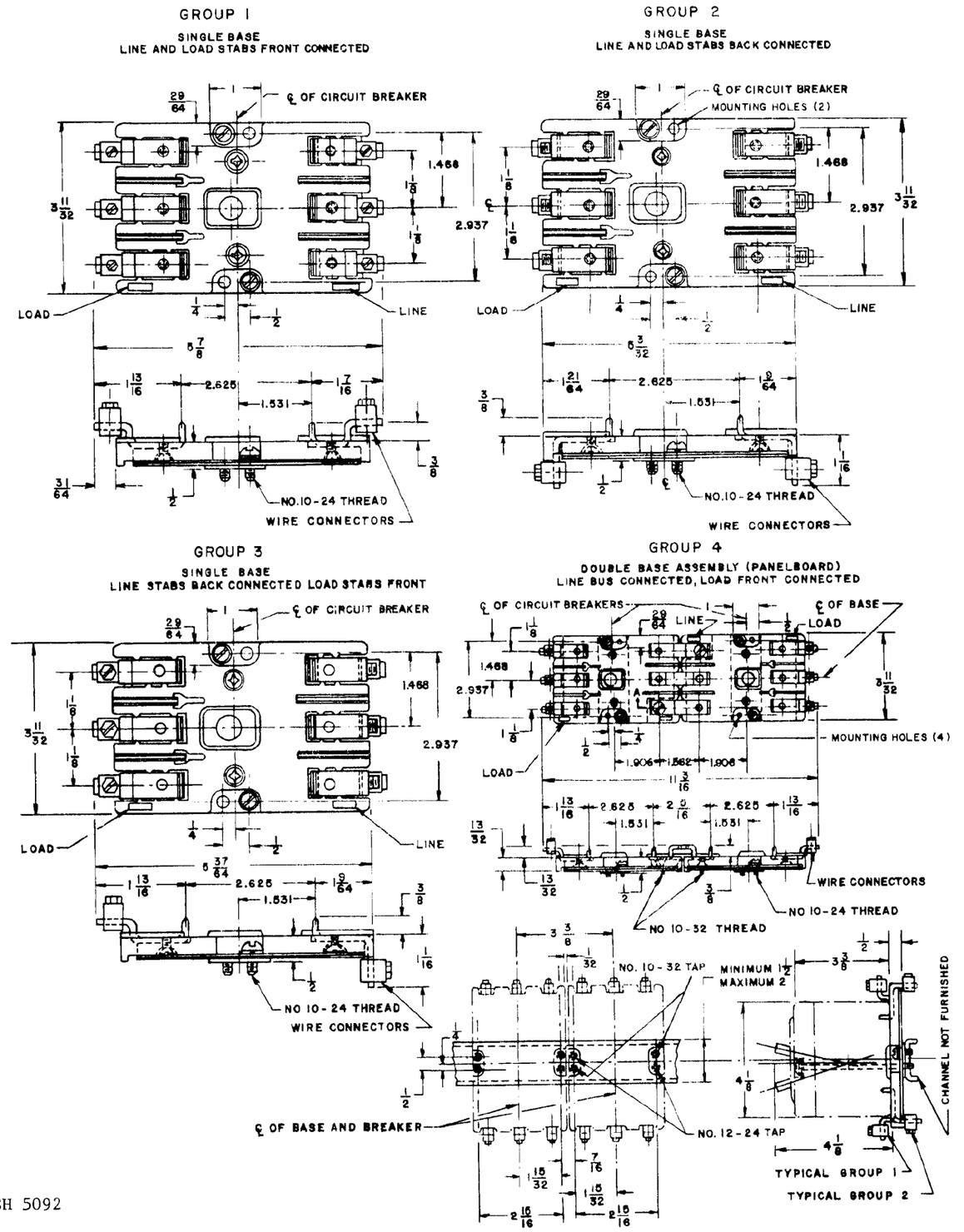


FIGURE 2. Mounting bases for circuit breaker type AQB-A50 and NQB-A50 outline and mounting dimensions.

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REQUIREMENTS:

Dimensions: See figures 1 and 2.
Handle boot: See figure 3 and table I.
Rated continuous current: 50 amperes.
Voltage and frequency: 500 Vat, 60 Hz.
500 Vat, 400 Hz.
250 Vdc.
Interrupting rating: 5,000 amperes (60 Hz at).
5,000 amperes (400 Hz at).
2,500 amperes (250 Vdc).
Peak let-through current: Not applicable.
Trip unit: Noninterchangeable .
Trip unit rating (amperes): See table II.
Protection: LTD and instantaneous, outer poles only,
LTD calibration: See table III.
Instantaneous calibration: See table II.
Current limiting fuse: Not applicable.
Type of connections: Back (removable mtg.).
Electric operator: Not available.
Undervoltage trip or shunt trip: Neither available.
Auxiliary switch: Not available.
Handle lock: Group 1, (Figure 1 of MIL-C-17361)
Additional identification: Front cover - “TRIPPED-RESET” and “ON-OFF”.
Frame - ampere rating.
Additional mounting requirements: Each circuit breaker shall have two threaded metal inserts molded in the cover for insertion of the supporting screws. Stud terminals or lug terminals connected to the circuit breaker shall not turn to reduce the electrical clearances. Lug terminals shall be supplied with mounting bases.
Shock test: Mounting shall be in a vertical position in accordance with fixture 6E of MIL-S-901.
Electrical test: The rate of operation shall be 6 operations per minute for a total of 6000 close-open operations.
Temperature rise tests: See table IV.

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TABLE I. Rubber handle boot.

Properties	
Color	Clear transparent
Specific gravity at 77°F	1.15 ± 0.02
Chemical classification	ZZ-R-765
Shore A hardness	50 ± 5
Minimum tensile strength (lb/in^2)	1100
Minimum elongation (percent)	450
Minimum tear strength (lb/in^2)	150
Minimum bayshore resiliency (percent)	40
Fatigue life at 125 percent strength (flexes)	800,000
Test requirements	
Color	1/
Shore A hardness	ASTM D 2240
Tensile strength	ASTM D 412
Elongation	ASTM D 412
Tear strength	ASTM D 624
Bayshore resiliency	ASTM D 412
Fatigue life	ASTM D 412

1/ A 0.075 inch slab of the rubber shall be placed in direct contact with a piece of newsprint. The rubber shall not interfere with the ability to read the newsprint.

TABLE II. Instantaneous trip calibrations.

Type designation	Trip element rating (amperes)	Test current (amperes)	Time (seconds)	Test current (amperes)	Maximum time 1/ (seconds)
AQB-A50	10,15,20 ac	30 times element rating	0.01 - 0.16	800	0.030
	25, 30 ac	600	0.015 - 0.33	800	0.33
	35,40,50 ac	600	0.30 - 1.0	800	0.80
	10,15,20 ac	40 times element rating	0.010 - 0.10	1200	0.08
	25,30 dc	900	0.012 - 0.16	1200	0.16
	35,40,50 dc	900	0.016 - 0.33	1200	0.30

1/ For qualification testing only, not for routine calibrations.

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TABLE III. Tripping characteristics - long time delay.

LTD trip element percent current	Tripping time 60 Hz ac and 400 Hz ac trip element dc trip element
115	Not less than 1 hour
138	Less than 1 hour
200	Between 10 and 200 seconds

TABLE IV. Size of cable and bus bars for temperature rise tests.

Copper rating (amperes)	Cable size		Copper bus size (inch)
	Ac	Dc	Ac and Dc
50	AWG 7	AWG 9	1/2 x1/8

Preparing activity:
Navy - SH
(Project 5925-N103-01)

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